DRG. NO. FT-001 · SHEET 1 / 5

FRANCIS LAB

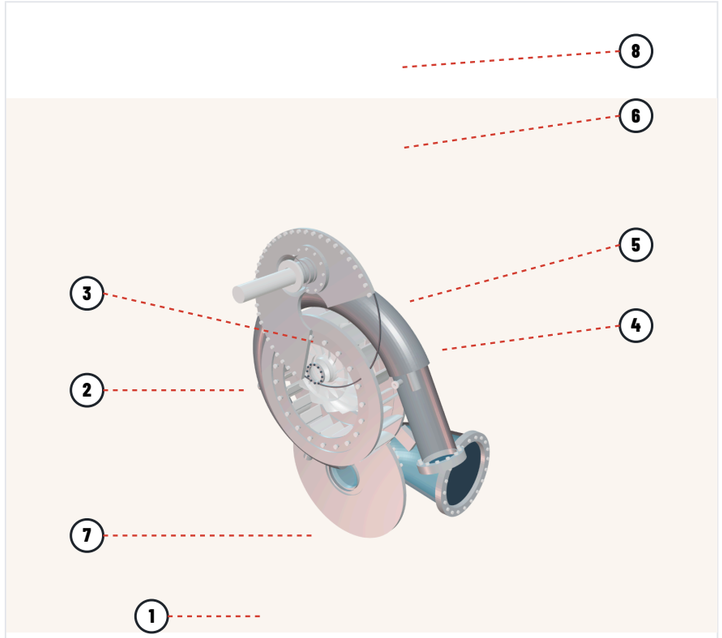
FRANCIS TURBINE APPARATUS · COEP TECH

SECTION A · THE APPARATUS

The Francis turbine, taken apart

The reaction hydraulic turbine in the lab, drawn schematically and exploded into its main parts. Numbers on each drawing match the list beside it.

FIG. 1A · REACTION TURBINE RIG EXPLODED · NTS



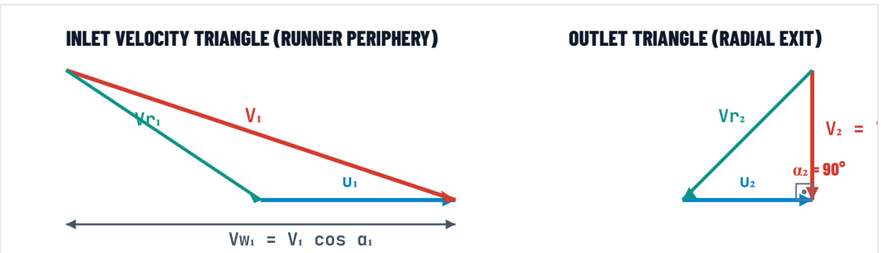
Francis Reaction Turbine

REACTION — Floor test rig · closed circuit

RUNNER	Mixed-flow · diameter [150] mm, [14] blades
HEAD	[10 to 30] m, set with centrifugal pump throttle
DRIVE PUMP	[5.5] kW monoblock pump, mains supplied
DISCHARGE	0 to [1500] L/min, Venturi-monitored

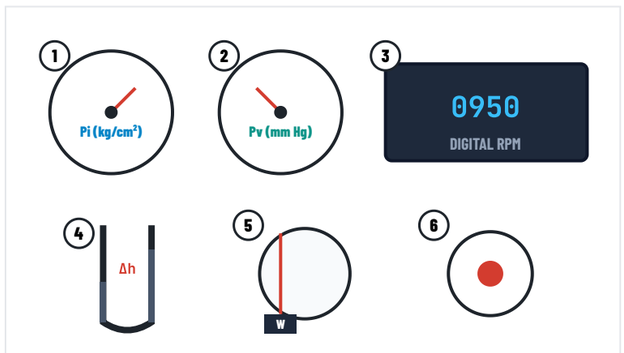
1. **Base frame & sump**· rigid steel channel bed, carries rig and recirculating tank
2. **Spiral casing**· decreasing volute cross-section maintains uniform flow velocity
3. **Stay ring**· stationary aerofoil vanes carry structural hydraulic casing load
4. **Wicket gates**· pivoted guide vanes regulate flow discharge and swirl angle α_1
5. **Francis runner**· converts pressure and angular momentum into shaft torque
6. **Turbine shaft**· precision ground alloy shaft supported on heavy-duty bearings
7. **Draft tube**· divergent elbow converts exit kinetic head into suction pressure
8. **Rope brake**· water-cooled brake drum, weights and spring balance apply load

FIG. 1B · THE VELOCITY TRIANGLES KINEMATICS



1. **Inlet whirl (Vw_1)**· carries total angular momentum into the runner periphery.
2. **Shockless entry**· blade angle β_1 matches Vr_1 to eliminate hydraulic separation loss.
3. **Radial discharge ($Vw_2 = 0$)**· leaves zero whirl kinetic energy waste at the exit.

RULE · EULER TURBINE LAW
Euler head is $H_e = (Vw_1 \cdot u_1 - Vw_2 \cdot u_2) / g$. For optimum efficiency, water leaves radially ($Vw_2 = 0$), yielding maximum energy conversion: $H_e = Vw_1 \cdot u_1 / g$.



1. **Delivery gauge**· reads spiral casing inlet pressure head Pi in kg/cm²
2. **Vacuum gauge**· reads draft tube suction vacuum Pv in mm of mercury
3. **Optical tachometer**· digital display reads turbine shaft speed N in rpm
4. **Venturi manometer**· U-tube differential mercury column measures flow rate Q
5. **Rope brake & weights**· hanger weights W and spring balance S quantify torque
6. **Handwheel & scale**· sets guide vane wicket gate opening from 0% to 100%

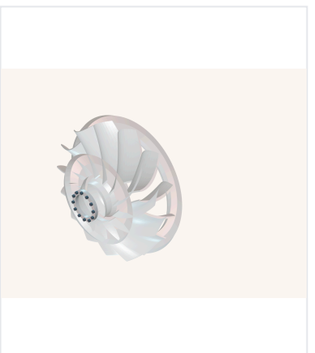


FIG. 1D · THE RUNNER ENERGY TRANSFER

Where energy converts

Water turns 90° through curved 3D runner channels, converting static pressure and dynamic swirl into shaft torque.

$$H_e = (Vw_1 \cdot u_1) / g$$

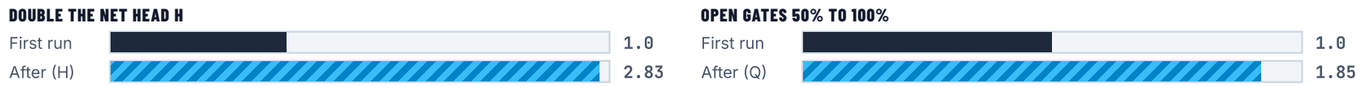
$$u_1 = \pi \cdot D_1 \cdot N / 60$$

$$\eta_h = (Vw_1 \cdot u_1) / (g \cdot H)$$

Runner dia [150] mm · Eye [80] mm · Max [1500] rpm

Change one thing

Bars show output power & flow compared to baseline design point (1.0). Hatched bars predicted from turbomachinery laws.



TITLE	SHEET	SCALE
THE FRANCIS TURBINE, TAKEN APART	1 / 5	NTS

Schematic drawings, not to scale. Bracketed red fields [___] are for your own apparatus values; check every part name against the real set-up before printing.

DRG. NO. FT-001 · SHEET 2 / 5

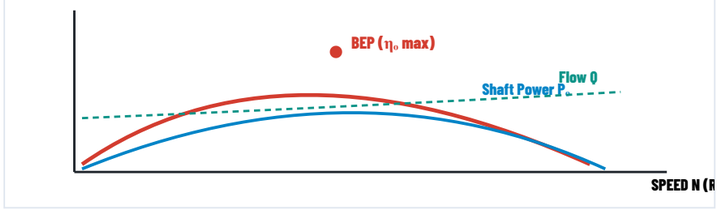
FRANCIS LAB

FRANCIS TURBINE APPARATUS · COEP TECH

SECTION B · THE EXPERIMENTS

Detail views: what we do with it

Six things the Francis turbine test rig teaches, from verifying the Euler equation to understanding cavitation, governing, and full characteristic curves.



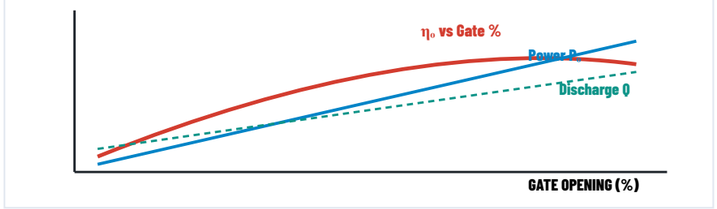
CARD A · HEAD · CONSTANT

Main characteristic curves

Keep net head H constant with pump throttle, and vary the brake load to sweep speed N from stall to runaway.

- Maintain casing pressure head and draft vacuum constant throughout the run.
- Increment dead weights on the rope brake hanger in 1 kg steps.
- Plot overall efficiency η₀, shaft power P₀, and discharge Q against speed N.
- Locate Best Efficiency Point (BEP) and design operating velocity ratio.

Group: [___] · Guide: [___]



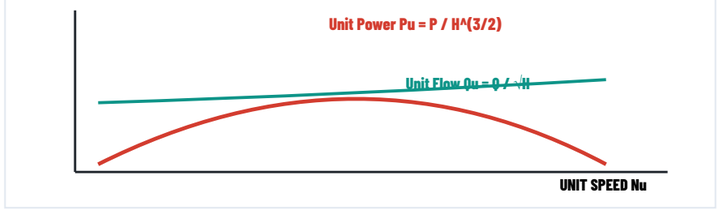
CARD B · SPEED · GOVERNING

How does gate opening govern power?

Hold turbine speed N constant at rated rpm by adjusting brake load as guide vane opening varies from 20% to 100%.

- Set wicket gates at 20%, 40%, 60%, 80%, and 100% opening positions.
- Balance brake load at each setting to keep shaft speed exactly at rated rpm.
- Plot efficiency η₀, discharge Q, and shaft power P₀ against gate opening percentage.
- Directly replicates hydro grid governor response during power dispatch.

Group: [___] · Guide: [___]



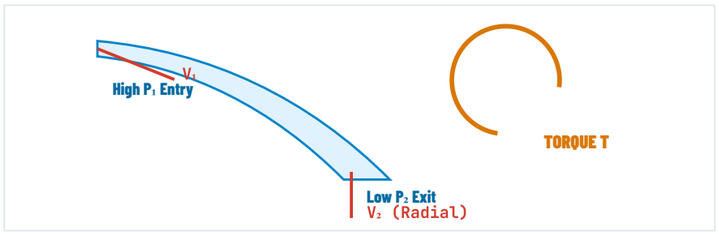
CARD C · SCALING · SIMILITUDE

Unit speed, unit flow, unit power

Collapse multi-head laboratory test runs onto unified dimensionless curves to predict performance under any reservoir head.

- Calculate Unit Speed Nu = N / √H, Flow Qu = Q / √H, and Power Pu = P / H^(3/2).
- Superimpose readings taken at 15 m, 20 m, and 25 m heads on one chart.
- A unified master envelope validates dynamic similarity and blade similitude.
- Standard industrial method for scaling scale-model data to full-size dams.

Group: [___] · Guide: [___]



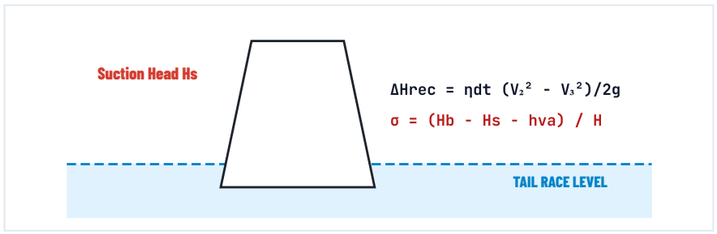
CARD D · MOMENTUM · EULER EQUATION

The energy exchange in the runner

Angular momentum flux entering at the runner rim must equal the shaft torque produced plus exit swirl.

- Measure inlet whirl $Vw_1 = V_1 \cos \alpha_1$ and calculate runner peripheral speed u_1.
- Calculate theoretical Euler head $H_e = (Vw_1 \cdot u_1 - Vw_2 \cdot u_2) / g$.
- Compare Euler head H_e with net manometric head H from gauges.
- The gap ($H - H_e$) quantifies hydraulic friction and blade passage loss.

Group: [___] · Guide: [___]



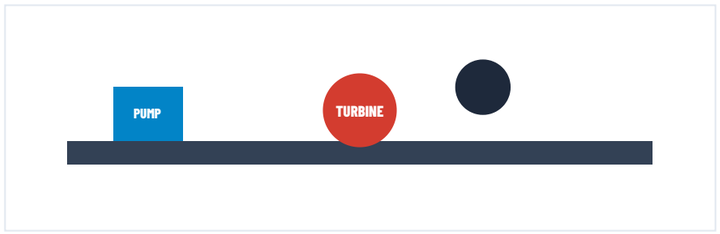
CARD E · DIFFUSER · CAVITATION

Why the draft tube saves energy

A divergent draft tube decelerates discharge flow, converting exit kinetic head back into suction pressure below the runner.

- Recovers exit kinetic energy: $\Delta Hrec = \eta dt \cdot (V_2^2 - V_3^2) / 2g$.
- Allows placing turbine above tail water by height Hs with zero head penalty.
- Calculate Thoma's cavitation factor: $\sigma = (Hb - Hs - hva) / H$.
- Prevents vapor pocket formation that causes blade pitting and destructive noise.

Group: [___] · Guide: [___]



CARD F · TEACHING · FLUID MACHINERY

The Francis turbine practical

Students run the test rig, tabulate manometric and brake readings, and plot performance characteristics.

- **Aim:** Evaluate overall efficiency and performance characteristics of Francis turbine.
- **Apparatus:** Francis turbine rig, supply pump, Venturimeter, rope brake, tachometer.
- **Readings:** Delivery Pi, vacuum Pv, manometer Δh, speed N, dead weight W, spring S.
- **Report:** Calculate heads, discharge, power, efficiency, and discuss energy losses.

Faculty: [___]

Experiment outlines follow the standard Fluid Machinery / Turbomachinery syllabus. Bracketed red fields [___] are for your group, guide and faculty names.

TITLE	SHEET	SCALE
DETAIL VIEWS: WHAT WE DO WITH IT	2 / 5	NTS

DRG. NO. FT-001 · SHEET 3 / 5

FRANCIS LAB

FRANCIS TURBINE APPARATUS · COEP TECH

SECTION C · THE THEORY

Why Francis turbines matter, and how we study them

High-pressure water gives up its energy inside a precision spiral casing and runner. Here is why that matters to engineers, and the route from head to shaft power.

C1 ENGINEERING CONTEXT

Why Francis turbines?

Over 60% of world hydropower is generated by Francis turbines. Here is why the reaction principle dominates global clean energy.

Baseload & grid stability
Instant black-start and fast spinning inertia balance transient grid fluctuations.

Pumped storage (PHS)
Reversible pump-turbines act as massive water batteries for solar/wind surplus.

Medium head mastery
Operates efficiently across 30 m to 600 m heads, unmatched by Pelton or Kaplan.

Peak 95%+ efficiency
Computerized CFD runner vane geometry captures virtually all fluid momentum.

Multi-purpose dams
Integrates seamlessly into municipal flood control and irrigation networks.

Teach the physics
Laboratory test rig demonstrates angular momentum and Euler reaction forces.

WHERE IT IS ALREADY USED

Koyna Hydro · Bhakra Nangal · Tehri Dam · Three Gorges · Grand Coulee · Itaipu · Pumped Storage · Micro-Hydro

Water enters the spiral casing with high pressure head. As it accelerates through wicket gates and turns axially through runner buckets, Euler work transfer converts both pressure and kinetic head into shaft torque.

C2 MATHEMATICAL FORMULATION

Revision table: five equations

The governing mathematical relationships from gauge pressure and Venturi flow to overall machine efficiency.

EQ A	EQ B	EQ C	EQ D	EQ E
$H = P_i/\rho g + P_v/\rho g + z$	$Q = C_d \cdot k \cdot \sqrt{(2gh)}$	$P_o = 2\pi NT / 60$	$P_{in} = \rho g Q H$	$\eta_o = P_o / P_{in}$
Net Head	Venturi Flow	Shaft Power	Water Power	Efficiency

FIVE STEPS TO UNDERSTANDING A REACTION TURBINE

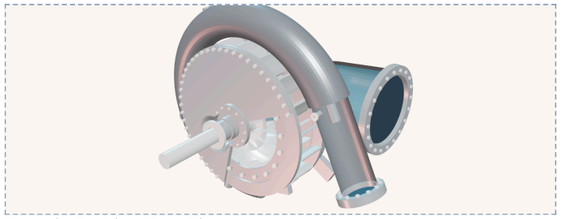
01	02	03	04	05
Guide it	**Accelerate**	**Extract**	**Diffuse**	**Deliver**
Wicket gates impart whirl velocity V_{w_1}.	Passage taper accelerates relative flow.	Runner vanes absorb angular momentum.	Draft tube recovers kinetic exit head.	Shaft delivers useful mechanical power.

FIVE-POINT PRACTICAL PROCEDURE

01	02	03	04	05
Prime rig	**Set head**	**Set gates**	**Load rig**	**Tabulate**
Check sump, prime pump, turn on drum cooling.	Open pump valve to establish test head H.	Lock wicket gates at test opening (e.g. 75%).	Hang weights on rope brake; allow N to settle.	Record Pi, Pv, Δh, N, W, S and compute η_o.

C3 COEP TECH

About the lab



Francis Turbine Test Rig
Fluid Mechanics & Turbomachinery Laboratory, Department of Mechanical Engineering, COEP Technological University, Pune.

1	3
FRANCIS RIG	CURVE REGIMES
$\eta \approx 75\%$	[1500]
PEAK RIG EFFICIENCY	MAX SPEED, RPM

Safety: Turn on brake drum cooling water before loading dynamometer to prevent rope burning. Keep clear of rotating shaft. Do not shut pump abruptly under full load.

Equations and applications follow the standard Theory of Machines / Fluid Machinery treatment. Bracketed red fields [___] are for your lab's own details.

TITLE	SHEET	SCALE
WHY FRANCIS TURBINES MATTER	3 / 5	NTS

DRG. NO. FT-001 · SHEET 4 / 5

SCAN • STEP INSIDE

FRANCIS TURBINE APPARATUS · HYDRAULIC MACHINERY LAB · COEP TECH

Two codes, two ways in. Open your phone camera and point it at either code. Nothing to install.

AR · CAMERA OVERLAY

Point your phone camera at the code.

CODE 1 · AR

The Francis turbine, on your bench

See the digital cutaway Francis turbine over your phone camera view of a real table, then explode it into runner, casing, and wicket gates.

01 Scan, then tap **Overlay on camera** .

02 Allow the camera and point it at a table.

03 Drag to turn it; use the **Explode slider**.

Works in a modern phone browser with a camera. Allow camera access when asked.

OR TYPE

`https://aryangadhe.github.io/ARVR/#ar`

WEB PAGE

Point your phone camera at the code.

CODE 2 · WEB

The whole lab, interactive

Take the Francis turbine apart in 3D, change the guide vane opening and the load, and watch the runner and telemetry respond.

01 Scan to open the page.

02 Drag the explode slider.

03 Change gate opening and brake load; compare with the formula.

Works in any modern browser on phones, tablets and laptops.

OR TYPE

`https://aryangadhe.github.io/ARVR/`

AR **Augmented reality**
The digital Francis turbine drawn over your phone camera view of a real table.

WEB **Web page**
Opens in any browser; explore the apparatus, velocity vectors and theory on any screen.

LAB **The real thing**
Scan, explore, then compare with the real Francis turbine on the bench.

Both codes open the lab's interactive page (FT-001). Test-scan before printing. Share the link so anyone can open it.

TITLE	SHEET	SCALE
SCAN · STEP INSIDE	4 / 5	NTS

DRG. NO. FT-001 · SHEET 5 / 5

HEAD • FLOW • POWER

FRANCIS TURBINE APPARATUS · FLUID MACHINERY LAB · COEP TECH

SECTION E · EXPERIMENT & CALCULATIONS

Run · Regulate · Compute

Two panels: how to run it, and how to work it out. Fill in the brackets from your own apparatus and verify turbine overall efficiency.

THE EXPERIMENT **STEP BY STEP**

Run it in six steps

01 **Check & Prime** · Ensure adequate water in sump, prime centrifugal pump, open brake cooling water, verify zero load on hanger.

02 **Start Pump** · Switch on pump motor with delivery valve closed; slowly open delivery throttle to pressurise spiral casing.

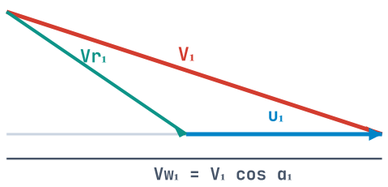
03 **Set Gate Opening** · Adjust wicket gate regulating handwheel to designated opening (e.g. 100%, 75%, 50% gate).

04 **Apply Brake Load** · Hang dead weights W on rope hanger; note spring reading S and steady speed N with tachometer.

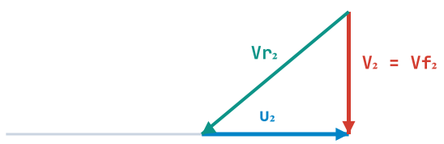
05 **Record Gauges** · Note casing pressure Pi, draft tube vacuum Pv, and mercury manometer deflection Δh across Venturimeter.

06 **Vary Load & Repeat** · Increment weights in uniform steps up to maximum load; repeat readings, then unload drum before stopping pump.

RUNNER INLET VELOCITY VECTORS



RUNNER OUTLET (RADIAL EXIT: $Vw_2 = 0$)



SAFETY: Ensure continuous cooling water to brake drum during all loaded runs. Never abruptly shut the delivery valve or supply pump motor under heavy load. Verify all electrical earthing before starting.

THE CALCULATION **WORKED EXAMPLE**

From readings to efficiency

Illustrative numbers: replace with your own apparatus readings.

TURBINE RIG CONSTANTS

Brake drum dia D = 0.28 m, rope d = 0.02 m
Effective radius Reff = (D+d)/2 = 0.15 m
Venturimeter d_1 = 100 mm, d_2 = 60 mm, Cd = 0.98

NET TOTAL HEAD (H)

Pi = 1.60 kg/cm² = 16.00 m of water
Pv = 150 mm Hg = 0.150 × 13.6 = 2.04 m
Datum z = 0.35 m → H = 18.39 m

DISCHARGE (Q) VIA VENTURIMETER

Manometer Δhm = 12.5 cm Hg = 0.125 m Hg
Head h = 0.125 × (13.6 - 1) = 1.575 m of water
Q = 0.0224 m³/s = 22.4 L/s

OUTPUT SHAFT POWER (P_o)

Load: W = 12.0 kg, S = 2.2 kg, N = 950 rpm
Torque T = (12.0 - 2.2) × 9.81 × 0.15 = 14.42 N·m
P_o = 2π × 950 × 14.42 / 60000 = 1.434 kW

WATER INPUT POWER (Pin)

Pin = (ρ · g · Q · H) / 1000
Pin = (1000 × 9.81 × 0.0224 × 18.39) / 1000
Pin = 4.041 kW

OVERALL EFFICIENCY (η_o)

η_o = (P_o / Pin) × 100%
η_o = (1.434 / 4.041) × 100%
η_o = 35.49%

YOUR OBSERVATION TABLE — RECORD READINGS DURING PRACTICAL

No.	Pi (kg/cm²)	Pv (mm Hg)	H (m)	Δh (cm Hg)	Q (L/s)	N (rpm)	W (kg)	S (kg)	(W-S) (kg)	T (N-m)	P_o (kW)	Pin (kW)	η_o (%)
1	1.60	150	18.39	12.5	22.4	950	12.0	2.2	9.8	14.42	1.434	4.041	35.49
2													
3													
4													
5													

H — Total Net Head
Effective net energy per unit weight available to the turbine: pressure head + draft suction vacuum + gauge elevation datum.

Q — Discharge
Volume flow rate entering the spiral casing, measured in real-time by the differential Venturimeter manometer.

η_o — Overall Efficiency
Ratio of useful mechanical brake shaft power to the total hydraulic energy supplied by the water flow: $\eta_o = P_o$ / Pin.

Worked example uses illustrative values. Replace the numbers with readings from your own apparatus and rotor.

TITLE	SHEET	SCALE
RUN · REGULATE · COMPUTE	5 / 5	NTS